TUGAS 04

Praktikum Bab 4 & Peta Konsepnya

Matakuliah Workshop DevOps

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Oleh :

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Politeknik Elektronika Negeri Surbaya

Departemen Teknik Informatika dan Komputer

Program Studi Teknik Informatika

September 2026

Praktikum 4

1. Direktori Kerja

```
toimul@DESKTOP-2OIAP1F:~$ mkdir -p ~/docker-lab/bab-4/{apache/sites,nginx/conf,certs,logs/nginx,app}
toimul@DESKTOP-2OIAP1F:~$ cd ~/docker-lab/bab-4
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ ls
apache  app  certs  logs  nginx
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ ls -R
.:
apache  app  certs  logs  nginx

./apache:
sites

./apache/sites:

./app:

./certs:

./logs:
nginx

./logs/nginx:

./nginx:
conf

./nginx/conf:
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$
```

2. Sertifikat TLS Laboratorium

```
./nginx/conf:
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
  -keyout certs/lab.key \
  -out certs/lab.crt \
  -subj "/CN=localhost/O=DevSecOps Docker Lab" \
  -addext "subjectAltName=DNS:localhost,IP:127.0.0.1"
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-----
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ chmod 600 certs/lab.key
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ chmod 644 certs/lab.crt
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$
```

3. File compose.yaml

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ nano compose.yaml
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$
```

```
  GNU nano 8.7.1                                                compose.yaml
services:
  proxy:
    image: nginx:alpine
    ports:
      - "127.0.0.1:8080:80"
      - "127.0.0.1:8443:443"
    volumes:
      - ./nginx/conf:/etc/nginx/conf.d:ro
      - ./certs:/etc/nginx/certs:ro
      - ./logs/nginx:/var/log/nginx
    networks:
      - web-net
    depends_on:
      apache-web:
        condition: service_started
      flask-app:
        condition: service_healthy

  apache-web:
    image: httpd:2.4-alpine
    volumes:
      - ./apache/sites:/usr/local/apache2/htdocs:ro
    networks:
      - web-net

  flask-app:
    build:
      context: ./app
    networks:
```

4. File Nginx/conf/default.conf

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ nano nginx/conf/default.conf
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$
```

```
  GNU nano 8.7.1                                        nginx/conf/default.conf
upstream apache_backend {
    server apache-web:80;
}

upstream flask_backend {
    server flask-app:5000;
}

server {
    listen 80;
    server_name localhost;
    return 301 https://$host:8443$request_uri;
}

server {
    listen 443 ssl;
    server_name localhost;

    ssl_certificate     /etc/nginx/certs/lab.crt;
    ssl_certificate_key /etc/nginx/certs/lab.key;
    ssl_protocols TLSv1.2 TLSv1.3;

    add_header X-Content-Type-Options "nosniff" always;
    add_header X-Frame-Options "DENY" always;
    add_header Referrer-Policy "no-referrer" always;

    location / {
        proxy_pass http://apache_backend;
        proxy_set_header Host $host;
```

5. File apache/sites/index.html

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ nano apache/sites/index.html
```

```
toimul@DESKTOP-2OIAP1F: ~
 GNU nano 8.7.1                                  apache/sites/index.html
<!DOCTYPE html>
<html lang="id">
<head>
    <meta charset="UTF-8">
    <title>DevSecOps Bab 4</title>
</head>
<body>
    <h1>Apache di Belakang Nginx</h1>
    <p>Halaman Bab 4 berhasil dilayani melalui reverse proxy.</p>
    <p><a href="/api/">Uji Flask API</a></p>
</body>
</html>
```

6. File app/requirements.txt

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ nano app/requirements.txt
```

```
toimul@DESKTOP-2OIAP1F: ~
 GNU nano 8.7.1                                  app/requirements
Flask==3.1.2
gunicorn==23.0.0
```

7. File app/app.py

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ nano app/app.py
```

```
 GNU nano 8.7.1                                  app/app.py *
from flask import Flask, jsonify, request

app = Flask(__name__)


@app.get("/")
def index():
    return jsonify(
        status="ok",
        service="flask-app",
        message="API Bab 4 berhasil diakses melalui Nginx",
        forwarded_proto=request.headers.get("X-Forwarded-Proto"),
    )


@app.get("/health")
def health():
    return jsonify(status="healthy"), 200
```

8. File app/Dockerfile

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ nano app/Dockerfile
```

```
toimul@DESKTOP-2OIAP1F: ~
  GNU nano 8.7.1                                          app/Dockerfile *
FROM python:3.12-slim

WORKDIR /app

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY app.py .

RUN useradd --system --uid 10001 --no-create-home appuser
USER appuser

EXPOSE 5000

CMD ["gunicorn", "--bind", "0.0.0.0:5000", "--workers", "2", "app:app"]
```

9. Validasi

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ cd ~/docker-lab/bab-4
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ find . -maxdepth 3 -type f | sort
./apache/sites/index.html
./app/Dockerfile
./app/app.py
./app/requirements.txt
./certs/lab.crt
./certs/lab.key
./compose.yaml
./nginx/conf/default.conf
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$
```

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ docker compose config
name: bab-4
services:
  apache-web:
    image: httpd:2.4-alpine
    networks:
      web-net: null
    volumes:
      - type: bind
        source: /home/toimul/docker-lab/bab-4/apache/sites
        target: /usr/local/apache2/htdocs
        read_only: true
        bind: {}
  flask-app:
    build:
      context: /home/toimul/docker-lab/bab-4/app
      dockerfile: Dockerfile
```

```
networks:
  web-net:
    name: bab-4_web-net
    driver: bridge
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ docker compose config --services
apache-web
flask-app
proxy
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$
```

10. Build dan Menjalankan Stack

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ docker compose up -d --build
[+] up 10/10
 ✔Image httpd:2.4-alpine Pulled
[+] Building 22.6s (14/14) FINISHED
 => [internal] load local bake definitions
 => => naming to docker.io/library/bab-4-flask-app:latest
[+] up 15/15king to docker.io/library/bab-4-flask-app:latest
 ✔Image httpd:2.4-alpine        Pulled
 ✔Image bab-4-flask-app         Built
 ✔Network bab-4_web-net         Created
 ✔Container bab-4-flask-app-1   Healthy
 ✔Container bab-4-apache-web-1  Started
 ✔Container bab-4-proxy-1       Started
```

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ docker compose ps
NAME                IMAGE              COMMAND                  SERVICE      CREATED         STATUS                   PORTS
bab-4-apache-web-1  httpd:2.4-alpine   "httpd-foreground"       apache-web   2 minutes ago   Up 2 minutes             80/tcp
bab-4-flask-app-1   bab-4-flask-app    "gunicorn --bind 0.0…"   flask-app    2 minutes ago   Up 2 minutes (healthy)   5000/tcp
bab-4-proxy-1       nginx:alpine       "/docker-entrypoint.…"   proxy        2 minutes ago   Up 2 minutes             127.0.0.1:8080
->80/tcp, 127.0.0.1:8443->443/tcp
```

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ docker compose logs --tail 100
proxy-1  | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
apache-web-1  | AH00558: httpd: Could not reliably determine the server's fully qualified domain name, using 172.18.0.3. Set the 'Ser
verName' directive globally to suppress this message
apache-web-1  | AH00558: httpd: Could not reliably determine the server's fully qualified domain name, using 172.18.0.3. Set the 'Ser
verName' directive globally to suppress this message
apache-web-1  | [Sat Oct 03 05:08:50.747778 2026] [mpm_event:notice] [pid 1:tid 1] AH00489: Apache/2.4.69 (Unix) configured -- resumi
ng normal operations
apache-web-1  | [Sat Oct 03 05:08:50.747863 2026] [core:notice] [pid 1:tid 1] AH00094: Command line: 'httpd -D FOREGROUND'
proxy-1       | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
proxy-1       | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
proxy-1       | 10-listen-on-ipv6-by-default.sh: info: can not modify /etc/nginx/conf.d/default.conf (read-only file system?)
proxy-1       | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
proxy-1       | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
proxy-1       | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
proxy-1       | /docker-entrypoint.sh: Configuration complete; ready for start up
flask-app-1   | [2026-10-03 05:08:50 +0000] [1] [INFO] Starting gunicorn 23.0.0
flask-app-1   | [2026-10-03 05:08:50 +0000] [1] [INFO] Listening at: http://0.0.0.0:5000 (1)
flask-app-1   | [2026-10-03 05:08:50 +0000] [1] [INFO] Using worker: sync
flask-app-1   | [2026-10-03 05:08:50 +0000] [7] [INFO] Booting worker with pid: 7
flask-app-1   | [2026-10-03 05:08:50 +0000] [8] [INFO] Booting worker with pid: 8
```

11. Pengujian Cepat

HTTP harus redirect ke HTTPS

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ curl -I http://localhost:8080/
HTTP/1.1 301 Moved Permanently
Server: nginx/1.31.5
Date: Sat, 03 Oct 2026 05:13:13 GMT
Content-Type: text/html
Content-Length: 169
Connection: keep-alive
Location: https://localhost:8443/
```

Halaman Apache melalui Nginx

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ curl -k -i https://localhost:8443/
HTTP/1.1 200 OK
Server: nginx/1.31.5
Date: Sat, 03 Oct 2026 05:14:02 GMT
Content-Type: text/html
Content-Length: 282
Connection: keep-alive
Last-Modified: Sat, 03 Oct 2026 05:01:41 GMT
ETag: "11a-65ce88b64f752"
Accept-Ranges: bytes
X-Content-Type-Options: nosniff
X-Frame-Options: DENY
Referrer-Policy: no-referrer

<!DOCTYPE html>
<html lang="id">
<head>
    <meta charset="UTF-8">
    <title>DevSecOps Bab 4</title>
</head>
<body>
    <h1>Apache di Belakang Nginx</h1>
    <p>Halaman Bab 4 berhasil dilayani melalui reverse proxy.</p>
    <p><a href="/api/">Uji Flask API</a></p>
</body>
</html>
```

Flask API melalui Nginx

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ curl -k https://localhost:8443/api/
{"forwarded_proto":"https","message":"API Bab 4 berhasil diakses melalui Nginx","service":"flask-app","status":"ok"}
```

Health endpoint Flask

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ curl -k -i https://localhost:8443/api/health
HTTP/1.1 200 OK
Server: nginx/1.31.5
Date: Sat, 03 Oct 2026 05:17:37 GMT
Content-Type: application/json
Content-Length: 21
Connection: keep-alive
X-Content-Type-Options: nosniff
X-Frame-Options: DENY
Referrer-Policy: no-referrer

{"status":"healthy"}
```

Informasi TLS

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ openssl s_client \
  -connect localhost:8443 \
  -servername localhost \
  -brief </dev/null
Connecting to 127.0.0.1
depth=0 CN=localhost, O=DevSecOps Docker Lab
verify error:num=18:self-signed certificate
CONNECTION ESTABLISHED
Protocol version: TLSv1.3
Ciphersuite: TLS_AES_256_GCM_SHA384
Peer certificate: CN=localhost, O=DevSecOps Docker Lab
Hash used: SHA256
Signature type: rsa_pss_rsae_sha256
Verification error: self-signed certificate
Negotiated TLS1.3 group: X25519MLKEM768
DONE
```

12. Pemeriksaan Isolasi

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ docker compose ps
NAME                 IMAGE              COMMAND                  SERVICE      CREATED          STATUS                    PORTS
bab-4-apache-web-1   httpd:2.4-alpine   "httpd-foreground"       apache-web   10 minutes ago   Up 10 minutes             80/tcp
bab-4-flask-app-1    bab-4-flask-app    "gunicorn --bind 0.0…"   flask-app    10 minutes ago   Up 10 minutes (healthy)   5000/tcp
bab-4-proxy-1        nginx:alpine       "/docker-entrypoint.…"   proxy        10 minutes ago   Up 10 minutes             127.0.0.1:80
80->80/tcp, 127.0.0.1:8443->443/tcp
```

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ docker network inspect bab-4_web-net
[
    {
        "Name": "bab-4_web-net",
        "Id": "1e6e38004c0ae16717c3a3f22edd7dc782730a9c6f5a4dec4270f43ca6e47921",
        "Created": "2026-10-03T12:08:50.047114059+07:00",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": null,
            "Config": [
                {
                    "Subnet": "172.18.0.0/16",
```

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ docker compose exec proxy wget -qO- http://apache-web/
<!DOCTYPE html>
<html lang="id">
<head>
    <meta charset="UTF-8">
    <title>DevSecOps Bab 4</title>
</head>
<body>
    <h1>Apache di Belakang Nginx</h1>
    <p>Halaman Bab 4 berhasil dilayani melalui reverse proxy.</p>
    <p><a href="/api/">Uji Flask API</a></p>
</body>
</html>
```

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ docker compose exec proxy wget -qO- http://flask-app:5000/health
{"status":"healthy"}
```

13. Log

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ docker compose logs proxy
proxy-1  | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
proxy-1  | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
proxy-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
proxy-1  | 10-listen-on-ipv6-by-default.sh: info: can not modify /etc/nginx/conf.d/default.conf (read-only file system?)
proxy-1  | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
proxy-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
proxy-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
proxy-1  | /docker-entrypoint.sh: Configuration complete; ready for start up
```

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ docker compose logs apache-web
apache-web-1  | AH00558: httpd: Could not reliably determine the server's fully qualified domain name, using 172.18.0.3. Set the 'Ser
verName' directive globally to suppress this message
apache-web-1  | AH00558: httpd: Could not reliably determine the server's fully qualified domain name, using 172.18.0.3. Set the 'Ser
verName' directive globally to suppress this message
apache-web-1  | [Sat Oct 03 05:08:50.747778 2026] [mpm_event:notice] [pid 1:tid 1] AH00489: Apache/2.4.69 (Unix) configured -- resumi
ng normal operations
apache-web-1  | [Sat Oct 03 05:08:50.747863 2026] [core:notice] [pid 1:tid 1] AH00094: Command line: 'httpd -D FOREGROUND'
apache-web-1  | 172.18.0.4 - - [03/Oct/2026:05:14:02 +0000] "GET / HTTP/1.1" 200 282
apache-web-1  | 172.18.0.4 - - [03/Oct/2026:05:20:06 +0000] "GET / HTTP/1.1" 200 282
```

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ docker compose logs flask-app
flask-app-1  | [2026-10-03 05:08:50 +0000] [1] [INFO] Starting gunicorn 23.0.0
flask-app-1  | [2026-10-03 05:08:50 +0000] [1] [INFO] Listening at: http://0.0.0.0:5000 (1)
flask-app-1  | [2026-10-03 05:08:50 +0000] [1] [INFO] Using worker: sync
flask-app-1  | [2026-10-03 05:08:50 +0000] [7] [INFO] Booting worker with pid: 7
flask-app-1  | [2026-10-03 05:08:50 +0000] [8] [INFO] Booting worker with pid: 8
```

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ tail -n 20 logs/nginx/access.log
172.18.0.1 - - [03/Oct/2026:05:13:13 +0000] "HEAD / HTTP/1.1" 301 0 "-" "curl/8.18.0" "-"
172.18.0.1 - - [03/Oct/2026:05:14:02 +0000] "GET / HTTP/1.1" 200 282 "-" "curl/8.18.0" "-"
172.18.0.1 - - [03/Oct/2026:05:17:12 +0000] "GET /api/ HTTP/1.1" 200 117 "-" "curl/8.18.0" "-"
172.18.0.1 - - [03/Oct/2026:05:17:37 +0000] "GET /api/health HTTP/1.1" 200 21 "-" "curl/8.18.0" "-"
```

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ tail -n 20 logs/nginx/error.log
2026/10/03 05:08:55 [notice] 1#1: using the "epoll" event method
2026/10/03 05:08:55 [notice] 1#1: nginx/1.31.5
2026/10/03 05:08:55 [notice] 1#1: built by gcc 15.2.0 (Alpine 15.2.0)
2026/10/03 05:08:55 [notice] 1#1: OS: Linux 6.18.33.2-microsoft-standard-WSL2
2026/10/03 05:08:55 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:1048576
2026/10/03 05:08:55 [notice] 1#1: start worker processes
2026/10/03 05:08:55 [notice] 1#1: start worker process 21
2026/10/03 05:08:55 [notice] 1#1: start worker process 22
2026/10/03 05:08:55 [notice] 1#1: start worker process 23
2026/10/03 05:08:55 [notice] 1#1: start worker process 24
2026/10/03 05:08:55 [notice] 1#1: start worker process 25
2026/10/03 05:08:55 [notice] 1#1: start worker process 26
2026/10/03 05:08:55 [notice] 1#1: start worker process 27
2026/10/03 05:08:55 [notice] 1#1: start worker process 28
```

15. Cleanup

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ docker compose down
[+] down 4/4
 ✔Container bab-4-proxy-1      Removed
 ✔Container bab-4-apache-web-1 Removed
 ✔Container bab-4-flask-app-1  Removed
 ✔Network bab-4_web-net        Removed
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$
```

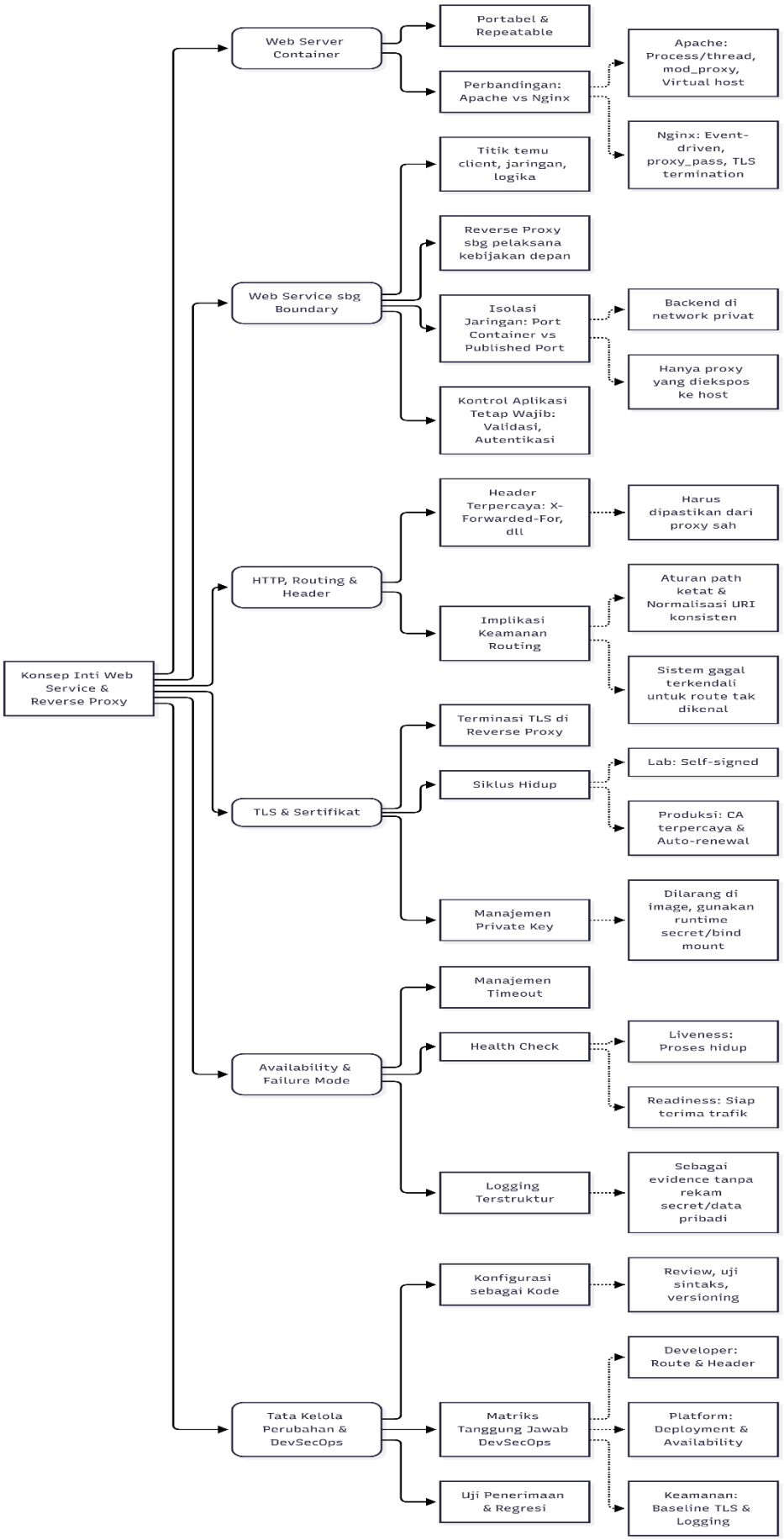
Beserta image local yang dihapus

```
toimul@DESKTOP-2OIAP1F:~/docker-lab/bab-4$ docker compose down --rmi local
[+] down 2/2
 ✔Dangling images              Removed
 ✔Image bab-4-flask-app:latest Removed
```

Checklist Pass

- [✔] docker compose config tidak menghasilkan galat: Pada bagian "9. Validasi", perintah docker compose config berhasil menampilkan struktur konfigurasi YAML tanpa ada pesan error.
- [✔] flask-app berstatus healthy: Pada bagian "10. Build dan Menjalankan Stack", hasil output dari docker compose ps dengan jelas menampilkan status Up 2 minutes (healthy) untuk service flask-app.
- [✔] HTTP port 8080 menghasilkan redirect 301: Pada bagian "11. Pengujian Cepat", eksekusi curl -I http://localhost:8080/ berhasil mendapatkan respons HTTP/1.1 301 Moved Permanently.
- [✔] HTTPS port 8443 menampilkan halaman Apache: Pada pengujian curl -k -i https://localhost:8443/, respons menampilkan status 200 OK beserta kode HTML yang berisi <h1>Apache di Belakang Nginx</h1>.
- [✔] /api/ menghasilkan JSON Flask: Pengujian ke endpoint https://localhost:8443/api/ berhasil mengembalikan respons berformat JSON yang berisi pesan "API Bab 4 berhasil diakses melalui Nginx".
- [✔] TLS 1.2 atau TLS 1.3 berhasil dinegosiasikan: Pada bagian "Informasi TLS", output dari perintah openssl s_client menunjukkan bahwa Protocol version: TLSv1.3 dan Negotiated TLS1.3 group berhasil digunakan.
- [✔] Apache dan Flask tidak memiliki published port: Pada bagian "12. Pemeriksaan Isolasi" (melalui docker compose ps), terlihat bahwa service apache-web (port 80/tcp) dan flask-app (port 5000/tcp) hanya bersifat internal, sedangkan published port ke host hanya dimiliki oleh service proxy.
- [✔] Request tercatat pada access log Nginx: Pada bagian "13. Log", perintah tail -n 20 logs/nginx/access.log menampilkan daftar request yang berhasil tercatat oleh Nginx, seperti request ke /, /api/, dan /api/health.

Peta Konsep


Konsep Inti Web Service & Reverse Proxy
Web Server Container
Portabel & Repeatable
Perbandingan: Apache vs Nginx
Apache: Process/thread, mod_proxy, Virtual host
Nginx: Event-driven, proxy_pass, TLS termination
Web Service sbg Boundary
Titik temu client, jaringan, logika
Reverse Proxy sbg pelaksana kebijakan depan
Isolasi Jaringan: Port Container vs Published Port
Backend di network privat
Hanya proxy yang diekspos ke host
Kontrol Aplikasi Tetap Wajib: Validasi, Autentikasi
HTTP, Routing & Header
Header Terpercaya: X-Forwarded-For, dll
Harus dipastikan dari proxy sah
Implikasi Keamanan Routing
Aturan path ketat & Normalisasi URI konsisten
Sistem gagal terkendali untuk route tak dikenal
TLS & Sertifikat
Terminasi TLS di Reverse Proxy
Siklus Hidup
Lab: Self-signed
Produksi: CA terpercaya & Auto-renewal
Manajemen Private Key
Dilarang di image, gunakan runtime secret/bind mount
Availability & Failure Mode
Manajemen Timeout
Health Check
Liveness: Proses hidup
Readiness: Siap terima trafik
Logging Terstruktur
Sebagai evidence tanpa rekam secret/data pribadi
Tata Kelola Perubahan & DevSecOps
Konfigurasi sebagai Kode
Review, uji sintaks, versioning
Matriks Tanggung Jawab DevSecOps
Developer: Route & Header
Platform: Deployment & Availability
Keamanan: Baseline TLS & Logging
Uji Penerimaan & Regresi

Penjelasan Peta Konsep

Berdasarkan teks tersebut, arsitektur web service dalam *container* dibangun di atas enam pilar utama:

1. Dasar Web Server Container

- Penggunaan *container* membuat konfigurasi aplikasi menjadi lebih portabel dan *repeatable*.
- Terdapat perbedaan arsitektur mendasar antara Apache (berbasis proses/thread per *request*, digunakan untuk *virtual host* dan kompatibilitas .htaccess) dengan Nginx (bersifat *event-driven asynchronous*, digunakan untuk *reverse proxy* dan terminasi TLS).
- *Reverse proxy* berfungsi untuk memisahkan *interface* publik dari *backend* internal.

2. Web Service sebagai Boundary Aplikasi

- *Web service* adalah titik pertemuan antara *client*, jaringan, dan logika aplikasi.
- *Reverse proxy* bertugas menerapkan kebijakan di lapisan depan (seperti terminasi TLS, pencatatan akses, dan pembatasan ukuran *request*).
- Meskipun menggunakan proksi, kontrol keamanan inti seperti validasi input dan autentikasi tetap harus berada di aplikasi *backend*.
- Terdapat pemisahan akses: hanya *reverse proxy* yang boleh memiliki *published port* ke *host*, sedangkan *backend* (seperti *database*) harus diisolasi di dalam *network* privat.

3. HTTP, Routing, dan Header Terpercaya

- *Reverse proxy* meneruskan informasi *client* menggunakan *header* (X-Forwarded-For, X-Forwarded-Proto, Host) yang hanya bisa dipercaya jika *request* terbukti datang dari proksi yang sah untuk menghindari manipulasi *header* oleh penyerang.
- Aturan *routing* tidak boleh longgar agar tidak membuka akses ke rute internal atau administratif.
- Sistem harus dirancang agar gagal secara terkendali apabila menerima rute yang tidak dikenal atau metode yang tidak didukung.

4. TLS dan Siklus Hidup Sertifikat

- Terminasi TLS dipusatkan pada *reverse proxy*.
- Sertifikat *self-signed* hanya digunakan untuk laboratorium, sedangkan di lingkungan produksi wajib menggunakan *Certificate Authority* (CA) terpercaya dan otomatisasi pembaruan.
- *Private key* dilarang keras dimasukkan ke dalam *image* atau repositori; kuncinya harus dipasang saat *runtime* sebagai *secret* atau *mount read-only*.

5. Availability, Health Check, dan Failure Mode

- *Reverse proxy* harus mengatur batas *timeout* yang tepat; tidak terlalu panjang agar tidak menguras *resource*, dan tidak terlalu pendek agar tidak menolak *request* valid.
- Status keandalan dibagi dua: *liveness* (proses aplikasi masih hidup) dan *readiness* (layanan sudah siap menerima trafik dan dependensi kritisnya terpenuhi).
- Log (baik akses maupun *error*) berfungsi sebagai *evidence* (bukti) dengan *timestamp* yang konsisten, namun tidak boleh merekam data rahasia (*secret*) atau data pribadi.

6. Implikasi Rekayasa dan Tata Kelola Perubahan (DevSecOps)

- Konfigurasi proksi harus diperlakukan sebagai kode yang melalui proses *review*, uji sintaks, dan pencatatan versi.
- Terdapat kolaborasi tanggung jawab DevSecOps: Developer (menentukan rute dan *header*), Platform (menjaga ketersediaan dan *deployment*), dan Keamanan (menentukan standar TLS dan *logging*).
- Pengujian penerimaan (*acceptance test*) dan pengujian regresi wajib diulang setiap kali ada perubahan dependensi atau *base image* untuk memastikan tidak ada perubahan perilaku HTTP yang tidak disengaja.